

Afterword The Mathematical Vision of Social Quantum Field Theory



7月 19, 2026

Afterword

The Mathematical Vision of Social Quantum Field Theory

"The history of mathematics is not merely the accumulation of theorems. It is the gradual discovery of deeper structures that reveal unexpected unity beneath apparent diversity."

1. Beyond Interdisciplinary Borrowing

Social Quantum Field Theory (SQFT) should not be understood merely as the application of concepts from quantum field theory to the social sciences.

Such an interpretation would underestimate its mathematical ambition.

The central objective of SQFT is to investigate whether a broad class of complex relational systems can be described using a common mathematical architecture built upon

- variational principles,
- field representations,
- symmetry,
- topology,
- information geometry,
- operator methods,
- category theory.

In this sense, SQFT is concerned less with transferring terminology than with identifying structural correspondences across mathematical disciplines.

2. Relations as Primitive Mathematical Objects

Much of modern mathematics begins with objects.

Relations are then introduced as secondary constructs connecting those objects.

SQFT proposes the opposite viewpoint.

Relations themselves become primitive.

Objects emerge only as stable localized structures within an evolving relational field.

Symbolically,

$$\text{Relations} \Rightarrow \text{Fields} \Rightarrow \text{Structures} \Rightarrow \text{Observables.} \quad (\text{A.1})$$

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This perspective aligns naturally with numerous developments in contemporary mathematics, including category theory, network science, information geometry, and higher algebra.

3. Mathematical Economy

One recurring principle throughout this monograph has been mathematical economy.

Whenever several mathematical descriptions exist,

the preferred formulation is the one requiring the fewest independent assumptions while preserving the greatest number of structural invariants.

This principle explains the repeated appearance of

- symmetry,
- conservation laws,
- variational formulations,
- topological invariants,
- effective theories.

Rather than isolated techniques,

these are viewed as manifestations of a common mathematical philosophy.

4. A Framework Rather Than a Final Theory

Every mathematical framework evolves.

Euclidean geometry expanded into differential geometry.

Newtonian mechanics evolved into Hamiltonian mechanics.

Classical probability developed into measure theory.

Likewise,

the present formulation of SQFT should be regarded as Version 1 of an evolving mathematical program.

Future researchers may

- simplify axioms,
- strengthen proofs,
- replace definitions,
- discover counterexamples,
- propose entirely new formulations.

Such developments represent progress rather than failure.

5. Mathematical Responsibility

The introduction of a new mathematical framework carries an important responsibility.

Novel terminology alone does not constitute scientific progress.

A useful theory must eventually satisfy demanding standards of rigor.

These include

- precise definitions,
- logical consistency,
- reproducible derivations,
- computational feasibility,
- empirical applicability.

Throughout this work,

care has been taken to distinguish

- proven theorems,
- working hypotheses,
- open conjectures,
- philosophical interpretations.

Maintaining these distinctions is essential for the continued development of SQFT.

6. Future Generations

No mathematical theory is completed by its first author.

The most influential theories have always been transformed by later generations.

Hilbert did not finish functional analysis.

Einstein did not complete modern differential geometry.

Grothendieck did not exhaust category theory.

Likewise,

if SQFT possesses lasting value,

its most important developments will likely be made by researchers who reinterpret, extend, simplify, or even correct the ideas presented here.

The role of this monograph is therefore foundational rather than definitive.

7. Closing Thought

Throughout this volume,

the recurring theme has been that complexity often arises not from the number of entities involved,

but from the richness of the relationships among them.

Mathematics provides a language capable of expressing those relationships with precision.

Whether the framework developed here ultimately becomes widely adopted remains uncertain.

What is certain, however, is that the search for deeper mathematical structures connecting diverse complex systems will continue.

If this work contributes even modestly to that search,

then it has fulfilled its purpose.

Structure	→Symmetry
	→Dynamics
	→Emergence
	→Understanding

"The value of a mathematical theory is measured not only by the problems it solves, but also by the new questions it enables us to ask."

Colophon

Title: *Social Quantum Field Theory: Toward a Mathematical Theory of Social Fields*

Author: Chou I-Hsien

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Language: English

Discipline: Mathematical Foundations of Complex Relational Systems

Keywords: Social Quantum Field Theory, Relational Fields, Variational Principles, Gauge Theory, Information Geometry, Topology, Tensor Networks, Category Theory, Complex Systems, Mathematical Modeling



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Foundational Charter of the Social Quantum Field Theory Research Community

7月 19, 2026

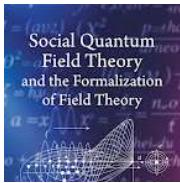
The SQFT Charter Foundational Charter of the Social Quantum Field Theory Research Community Preamble The Social Quantum Field Theory (SQFT) Research Community is founded upon the conviction that mathematics advances through open inquiry ...

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社會量子場論（SQFT）超越布迪厄的場域理論，建構關係社會學的形式化 1~10

7月 15, 2026

預印本 社會量子場論與場論的形式化 周易先 獨立研究者 · 版本 2.0（數學版） · 2026年7月
摘要 本文提出社會量子場論（SQFT）作為用以表示複雜社會系統中關係動態的形式數學框



型。SQFT並不主張社會體系遵守量子力學法則。它採用量子力學法則來解釋社會現象。...

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9月 27, 2025



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